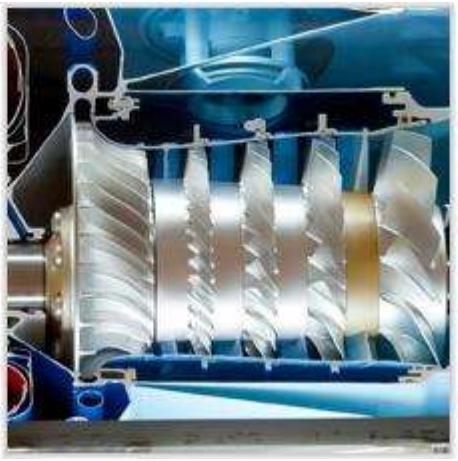
Description

Image


Caption

1. Jet engine. © Rolls-Royce plc 2. Cross section of a gas turbine showing its titanium blades. © iStockphoto 3. Titanium cladding for buildings. © John Fernandez

The material

Titan was a Greek god, remarkable for his size and strength. His name has been appropriated many times, not always aptly (think of the Titanic). But the alloys of titanium merit the association: the strongest of them have the highest strength-to-weight ratio of any structural metal, about 25% greater than the best alloys of aluminum or steel. Titanium alloys can be used at temperatures up to 500 C - compressor blades of aircraft turbines are made of them. They have unusually poor thermal and electrical conductivity, and low expansion coefficients. The alloy Ti 6%Al 4%V is used in quantities that exceed those of all other titanium alloys combined. The data in this record describes it and similar alloys.

Compositional summary

Ti + alloying elements, e.g. Al, Zr, Cr, Mo, Si, Sn, Ni, Fe, V

General properties

Density		4.43e3	-	4.79e3	kg/m^3
Price	*	17.6	-	22.1	GBP/kg
Date first used		1952			

Mechanical properties

Young's modulus		110	-	120	GPa
Shear modulus		38	-	45	GPa
Bulk modulus		110	-	120	GPa
Poisson's ratio		0.33	-	0.35	
Yield strength (elastic limit)		701	-	1.09e3	MPa

Tensile strength		763	-	1.19e3	MPa
Compressive strength	*	680	-	1.15e3	MPa
Elongation		6	-	20	% strain
Hardness - Vickers		223	-	373	HV
Fatigue strength at 10^7 cycles	*	412	-	636	MPa
Fracture toughness		51.3	-	86.1	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.001	-	0.002	

Thermal properties

Melting point		1.5e3	-	1.66e3	°C
Maximum service temperature		380	-	487	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Poor conductor			
Thermal conductivity		6	-	8.99	W/m.°C
Specific heat capacity		534	-	544	J/kg.°C
Thermal expansion coefficient		8.27	-	9.8	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Poor conductor			
Electrical resistivity		130	-	190	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	Yes
Notes	Al (aluminum) and Ti (titanium) added to the 2018 US critical minerals list

Processability

Castability	3			
Formability	2	-	4	
Machinability	1	-	3	
Weldability	4	-	5	
Solder/brazability	1	-	2	

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Excellent
Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	

	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Excellent
Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Acceptable
Sulfuric acid (10%)	Acceptable
Sulfuric acid (70%)	Acceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Limited use
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent

Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Limited use
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Limited use

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Limited use
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	7.6e6	tonne/yr
Reserves, principal component	6.73e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	*	40.2	-	44.3	kg/kg
Embodied energy (virgin grade production)	*	654	-	721	MJ/kg
Water usage	*	187	-	207	l/kg

Material processing: energy

Metal casting, embodied energy	* 53.5	-	60.6	MJ/kg
Metal extrusion, hot, embodied energy	* 75.6	-	105	MJ/kg
Metal rolling and forging, embodied energy	* 38.5	-	53.3	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 84	-	99.7	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1.33e4	-	1.52e4	MJ/kg
Machining, coarse, embodied energy	* 25.8	-	31.7	MJ/kg
Machining, fine, embodied energy	* 48.3	-	63	MJ/kg
Machining, grinding, embodied energy	* 70.8	-	94.2	MJ/kg
Machining, laser, CO2, embodied energy	* 1.59	-	1.8	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.91	-	4.44	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 4.95	-	6.87	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 2.52	-	3.49	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 6.87	-	8.15	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 1.1e3	-	1.25e3	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.69	-	2.08	kg/kg
Machining, fine, climate change (CO2-eq)	* 3.16	-	4.13	kg/kg
Machining, grinding, climate change (CO2-eq)	* 4.63	-	6.17	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.131	-	0.148	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 11.7	-	12.8	kg/kg
Embodied energy, recycling	* 445	-	490	MJ/kg
Recycle fraction in current supply	49			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Extracting titanium from its ores is very energy intensive. It can be recycled provided it is not contaminated with oxygen.

Supporting information

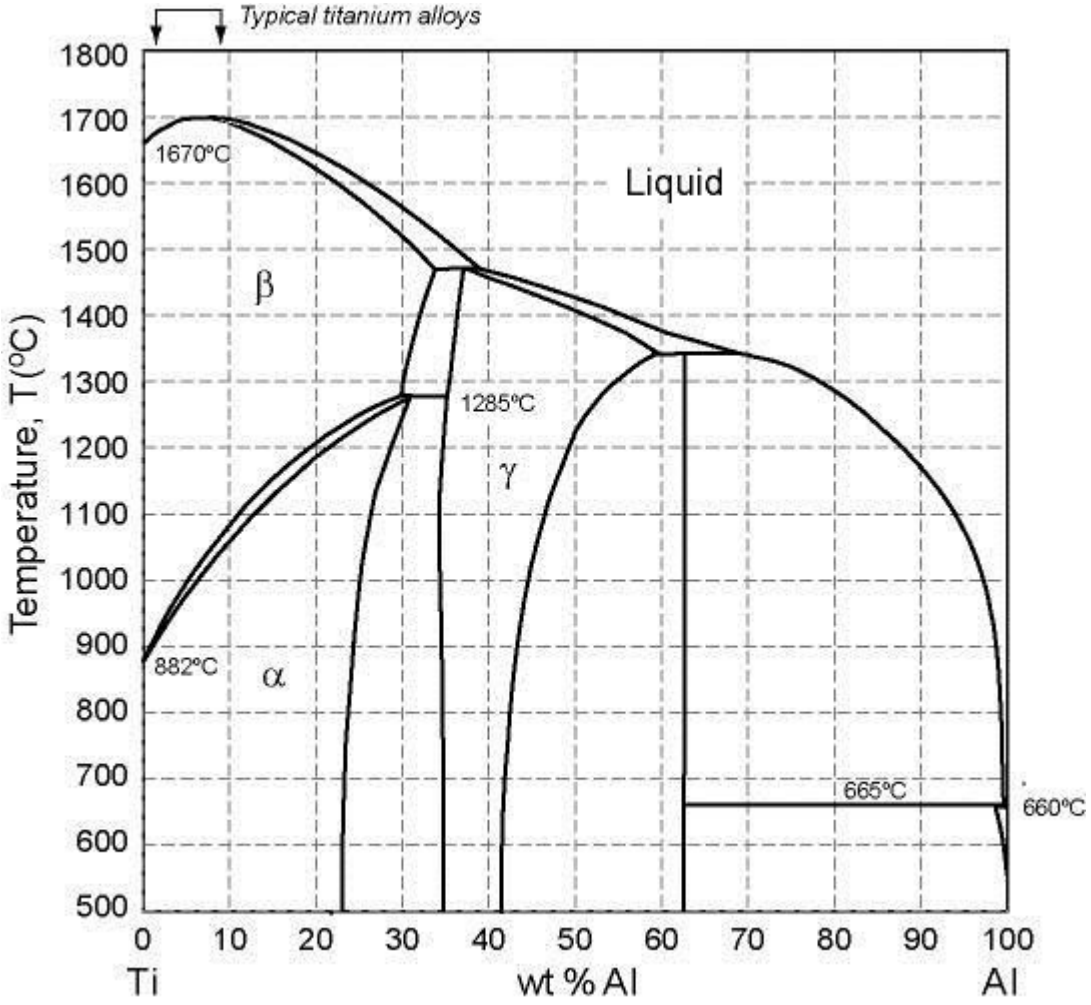
Design guidelines

Titanium alloys are expensive, requiring vacuum processing to prevent take up of oxygen, which makes them brittle. But they unusually strong, light and corrosion resistant, so much so that pure titanium can be implanted in the body to repair broken bones. More usually it is alloyed with aluminum and vanadium (Ti with 6%Al 4%V, or simply Ti - 6 - 4) to give a material that can be forged and worked yet has good resistance to creep. Titanium alloys have limited ductility - sheet cannot easily be bent to radii less than 1.5 times its thickness. They can - with difficulty - be welded, but are easy to diffusion bond.

Technical notes

There are four groups of titanium alloys: alpha alloys, near-alpha alloys, alpha-beta alloys, and beta alloys. The alpha alloys have an hcp crystal structure; the beta alloys are bcc. Alpha alloys are preferred for high temperature applications because of their creep resistance and for cryogenic applications because of their good toughness at low temperatures. A designation system with some logic to it simply lists the quantities of the principal alloying additions; thus 'Ti-8-1-1' contains 8% aluminum, 1% molybdenum and 1% vanadium; and 'Ti-6-4' means 6% aluminum and 4% vanadium. The alloy Ti 5% Al 2.5% Sn is the most widely used alpha alloy; it is used in space and aircraft structures. The alloy Ti 6% Al 4% V has a mixed alpha-beta structure; it is the most widely used of all titanium alloys.

Phase diagram



Phase diagram description

Titanium alloys have complex compositions. Most are based on titanium (Ti) with 2 - 8% aluminum (Al) for which this is the phase diagram, with additions of vanadium, tin, zirconium and molybdenum.

Typical uses

Aircraft turbine blades, general aerospace applications, chemical engineering, heat exchangers, bioengineering, medical, missile fuel tanks, heat exchangers, compressors, valve bodies, light springs, surgical implants, marine hardware, paper-pulp equipment, sports equipment such as golf clubs and bicycles, casings for mobile phones and portable computers.

Links

ProcessUniverse

Producers

Reference

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